The Intention-Execution Disconnect in Medical AI: The ReXecution Framework for Evaluating Real-World Clinical Performance

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Problem

Past Evaluation: Automatically Generated Tasks

AI-Generated Question: Did the patient show any abnormalities in the left hemidiaphragm in the same month after being diagnosed with unspecified pleural effusion 2 years ago? **Answer:** No.

Our Evaluation: Radiologist-Curated Tasks

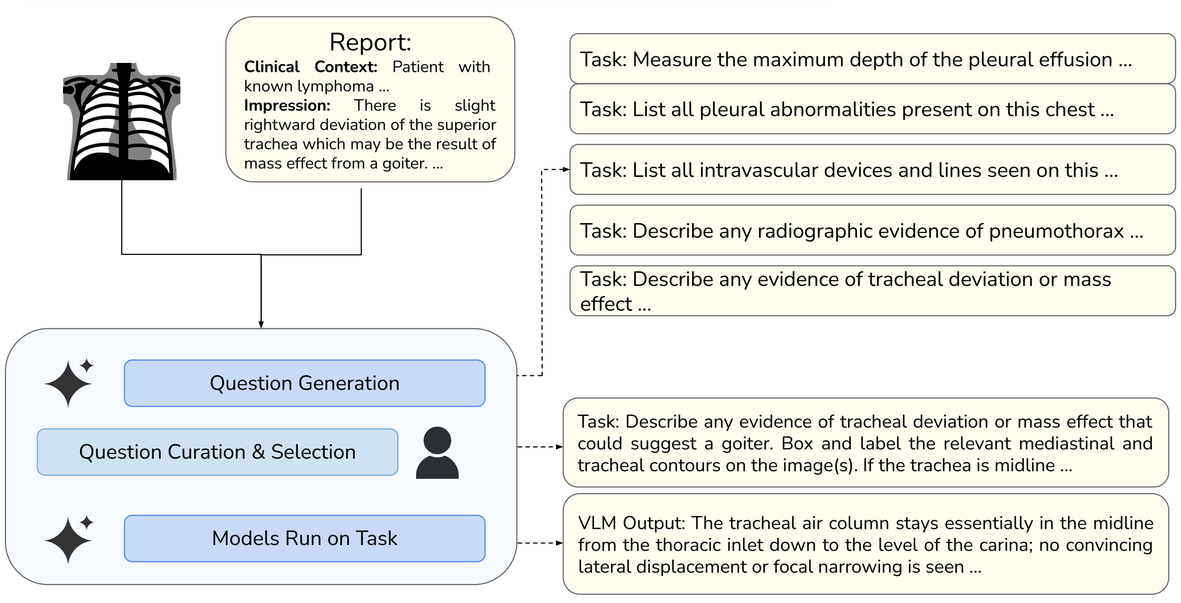
Evaluate this image for possible causes of bloody sputum and cough. Pose a differential diagnosis, list all evidence ...

- Large benchmarks for radiology vision-language models (VLMs) automatically generate and score thousands of question-answer pairs.[1,2,3]
- These benchmarks suggest that current VLMs are approaching clinician-level performance across many chest X-ray tasks.[1]
- However, AI-generated questions and answers are frequently **unrealistically simple, overly niche or otherwise irrelevant to clinical practice.**

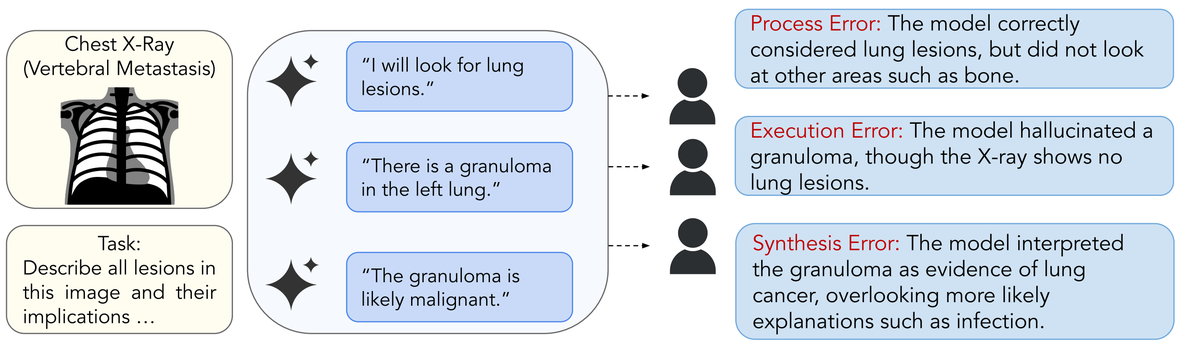
Question: Do radiologists find VLMs useful for realistic complex tasks from clinical workflows?

ReXscore Framework

LLM-Augmented Dataset Curation



- Using ChatGPT-o3, we generated 500 tasks based on 100 studies from the MIMIC-CXR test set (5 tasks/study).
- Radiologists selected 1 task per study, rewriting questions as needed to ensure realism, complexity and diversity.
- We tested ChatGPT-o3 (generalist, closed-source, text+image outputs) and MedGemma (medical specialist, open-source, text output) on the final 100 tasks.



While past benchmarks often feature easily scored short-answer/ multiple-choice questions, our complex tasks elicit long, detailed answers. For a nuanced assessment, we manually scored answers from 1-5 on:

- **Process.** Did the model correctly interpret the task and plan an appropriate strategy?
- **Execution.** Did it accurately read the image(s), without hallucinations or omissions?
- **Synthesis.** Did it draw appropriate conclusions from image findings?
- **Language.** Was the response clear, concise, and professional?
- **Image Content.** Were image outputs accurately annotated?
- **Image Style.** Were image outputs neatly composed?

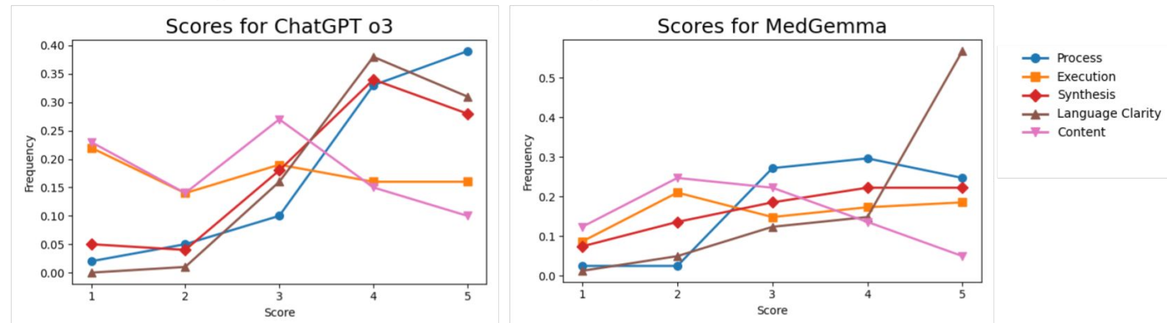
We also produced a summary "Content" score, equaling the minimum of Process, Execution and Synthesis scores. Answers only received scores of 5 on Content if they performed perfectly on Process, Execution and Synthesis.

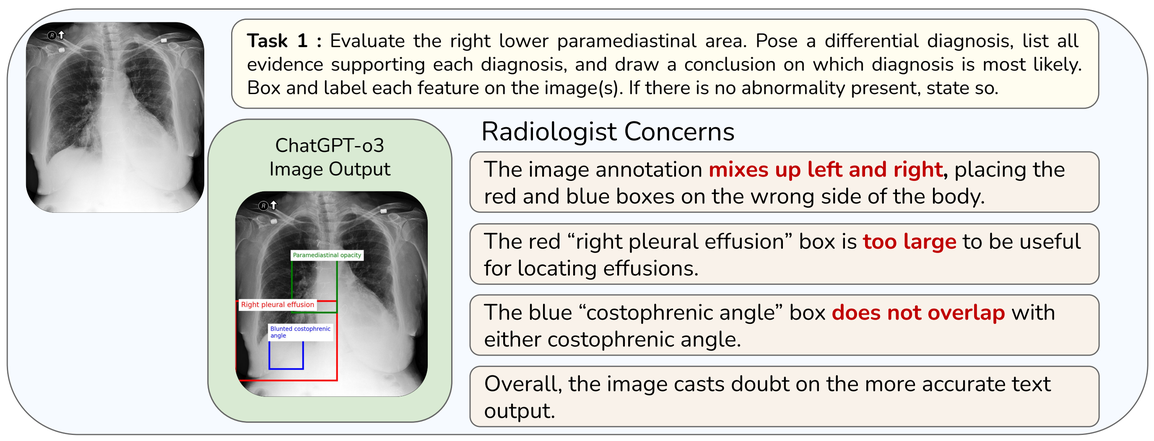
Results

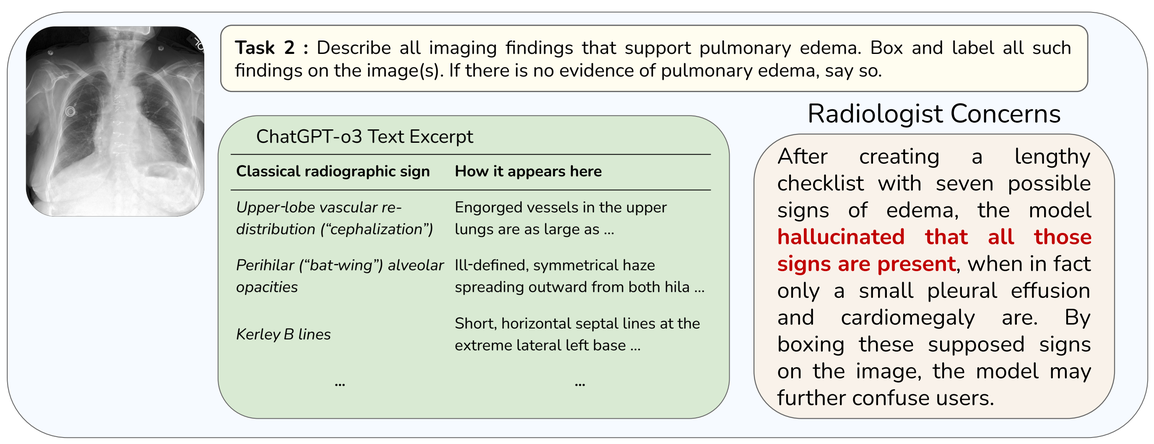
- ChatGPT-o3 received an average Content score of 2.71, with **only 10.0% of outputs receiving a 5.**
- MedGemma received an average Content score of 2.84, with **only 4.9% of outputs receiving a score of 5.**
- Both models tended to score well on Process and much worse on Execution.
- 36.2% of ChatGPT-o3's images received an Image Style score of 5, while only 10.1% received a 5 for Image Content.

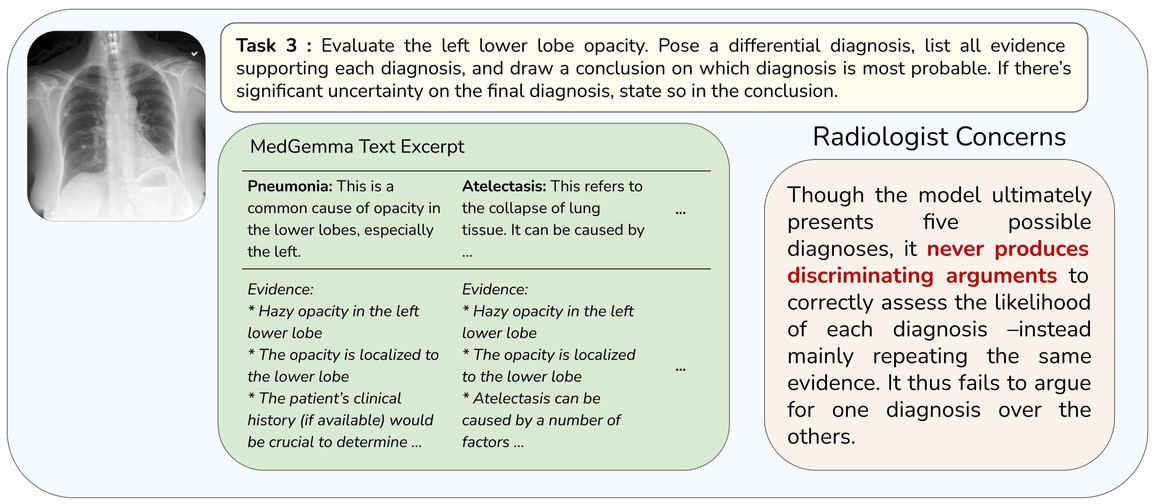
In the abstract, models showed strong medical knowledge and planning skills (e.g. when interpreting questions, breaking larger tasks down into subtasks, and choosing which details to illustrate in an image output)

In practice, they struggled to execute those plans due to errors in visual understanding, as seen in the following example tasks.









Conclusion

- By manually curating realistic, complex questions and scoring their answers, we find shortcomings in current models that limit their impact on real-world clinical workflows.
- Our findings complicate recent claims that high-performing VLMs are generally approaching clinician-level performance on chest X-ray tasks.

Limitations: Even with only 100 tasks, our dataset required intensive annotation by a resident and attending radiologist, limiting scale and reproducibility relative to past benchmarks.

References

1. A. Pal, et al. Rexvqa: A large-scale visual question answering benchmark for generalist chest x-ray understanding (2025).
2. S. Bae, et al. Mimic-ext-mimic-cxr-vqa: A complex, diverse, and large-scale visual question answering dataset for chest x-ray images (2024).
3. Liu, K. et al. Gemex: A large-scale, groundable, and explainable medical vqa benchmark for chest x-ray diagnosis (2025).